```
clc; clear; close all;
```

FIR FILTER PARAMETERS

```
b = [0.0625 0.2500 0.3750 0.2500 0.0625];
Fs = 1000;
N = 128;

x = [1 zeros(1,N-1)];
n = 0:N-1;
```

1. DIRECT REALIZATION

```
yd = filter(b,1,x);
```

2. CASCADE REALIZATION

```
z = roots(b);
g = b(1);
yc = g*x;

for k = 1:length(z)
    yc = filter([1 -z(k)],1,yc);
end

yc = real(yc);
```

3. SYMMETRY AND LINEAR PHASE

```
symErr = max(abs(b-fliplr(b)));
D = (length(b)-1)/2;

if symErr < 1e-12
    symmetry = 'Symmetric';
else
    symmetry = 'Not Symmetric';
end
```

4. FREQUENCY, PHASE AND GROUP DELAY

```
[H,w] = freqz(b,1,1024);
phase = unwrap(angle(H));
gd = grpdelay(b,1,1024);
```

5. NUMERICAL COMPARISON

```
err = max(abs(yd-yc));
dev = err/max(abs(yd))*100;
```

==========

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DISPLAY RESULTS ==========

```
fprintf('\n=========================================\n');
fprintf('       FIR FILTER REALIZATION RESULTS\n');
fprintf('=========================================\n');

fprintf('\nFIR Coefficients:\n');
disp(b);

fprintf('Sampling Frequency = %d Hz\n',Fs);
fprintf('Number of Samples  = %d\n',N);

fprintf('Coefficient Symmetry = %s\n',symmetry);
fprintf('Symmetry Error       = %.3e\n',symErr);
fprintf('Group Delay          = %.2f samples\n',D);

fprintf('\nMaximum Error        = %.3e\n',err);
fprintf('Percentage Deviation = %.3e %%\n',dev);
```

```
=========================================
       FIR FILTER REALIZATION RESULTS
=========================================

FIR Coefficients:
    0.0625    0.2500    0.3750    0.2500    0.0625

Sampling Frequency = 1000 Hz
Number of Samples  = 128
Coefficient Symmetry = Symmetric
Symmetry Error       = 0.000e+00
Group Delay          = 2.00 samples

Maximum Error        = 8.327e-16
Percentage Deviation = 2.220e-13 %
```

==========

==========

IMPULSE RESPONSE VALUES ==========

```
fprintf('\n=========================================\n');
fprintf('FIRST 10 IMPULSE RESPONSE VALUES\n');
```

```
fprintf('=========================================\n');

fprintf('   n       Direct        Cascade\n');

for k = 1:10
    fprintf('%4d    %10.6f    %10.6f\n', ...
        n(k),yd(k),yc(k));
end
```

```
=========================================
FIRST 10 IMPULSE RESPONSE VALUES
=========================================
   n       Direct        Cascade
   0      0.062500      0.062500
   1      0.250000      0.250000
   2      0.375000      0.375000
   3      0.250000      0.250000
   4      0.062500      0.062500
   5      0.000000      0.000000
   6      0.000000      0.000000
   7      0.000000      0.000000
   8      0.000000      0.000000
   9      0.000000      0.000000
```

==========

==========

CASCADE ZEROS ==========

```
fprintf('\n=========================================\n');
fprintf('CASCADE REALIZATION ZEROS\n');
fprintf('=========================================\n');

disp(z);
```

```
=========================================
CASCADE REALIZATION ZEROS
=========================================
  -1.0002 + 0.0000i
  -1.0000 + 0.0002i
  -1.0000 - 0.0002i
  -0.9998 + 0.0000i
```

==========

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OUTPUT PARAMETER TABLE ==========

```
Parameter = {
    'Direct realization output'
    'Cascade realization output'
    'Impulse response'
    'Phase response'
    'Comparison of realizations'
    };

Observation = {
    'Direct FIR filtering'
    'Cascade of first-order sections'
    'Equal to FIR coefficient sequence'
    'Linear phase'
    'Equivalent transfer-function behavior'
    };

Value = {
    sprintf('[%.4f %.4f %.4f %.4f %.4f]',yd(1:5))
    sprintf('[%.4f %.4f %.4f %.4f %.4f]',yc(1:5))
    sprintf('[%.4f %.4f %.4f %.4f %.4f]',yd(1:5))
    sprintf('Group delay = %.2f samples',D)
    sprintf('Max error = %.3e; Deviation = %.3e %%',err,dev)
    };

Unit_Note = {
    'Amplitude'
    'Amplitude'
    'h[n]'
    'Samples'
    'Numerical precision'
    };

Output_Table = table(Parameter,Observation,Value,Unit_Note);

fprintf('\n=============================================\n');
fprintf('OUTPUT PARAMETER TABLE\n');
fprintf('=============================================\n');

disp(Output_Table);
```

```
=============================================
OUTPUT PARAMETER TABLE
=============================================
              Parameter
Observation
Value                                    Unit_Note
    ___________________________________
_____________________________________________
_____________________________________________________    ____________________

    {'Direct realization output' }    {'Direct FIR
filtering'                  }    {'[0.0625 0.2500 0.3750 0.2500
0.0625]'          }    {'Amplitude'          }
```

```
    {'Cascade realization output'}    {'Cascade of first-order
sections'      }    {'[0.0625 0.2500 0.3750 0.2500 0.0625]'          }
{'Amplitude'          }
    {'Impulse response'          }    {'Equal to FIR coefficient
sequence'    }    {'[0.0625 0.2500 0.3750 0.2500 0.0625]'          }
{'h[n]'               }
    {'Phase response'            }    {'Linear
phase'                         }    {'Group delay = 2.00
samples'                    }    {'Samples'           }
    {'Comparison of realizations'}    {'Equivalent transfer-function
behavior'}    {'Max error = 8.327e-16; Deviation = 2.220e-13 %'}
{'Numerical precision'}
```

==========

==========

GRAPH 1: DIRECT VS CASCADE IMPULSE RESPONSE

==========

```
figure;

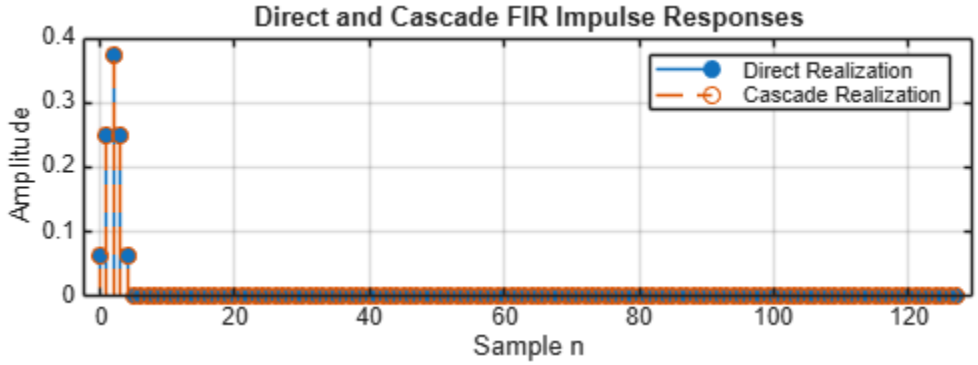
stem(n,yd,'filled');
hold on;
stem(n,yc,'--');

grid on;

xlabel('Sample n');
ylabel('Amplitude');

title('Direct and Cascade FIR Impulse Responses');

legend('Direct Realization','Cascade Realization');
```



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GRAPH 2: MAGNITUDE RESPONSE
==========

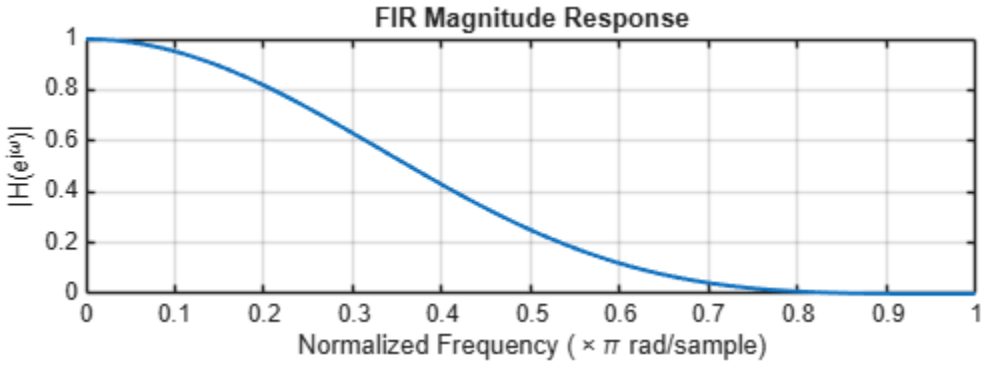
```
figure;

plot(w/pi,abs(H),'LineWidth',1.5);

grid on;

xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('|H(e^{j\omega})|');

title('FIR Magnitude Response');
```



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GRAPH 3: PHASE RESPONSE ==========

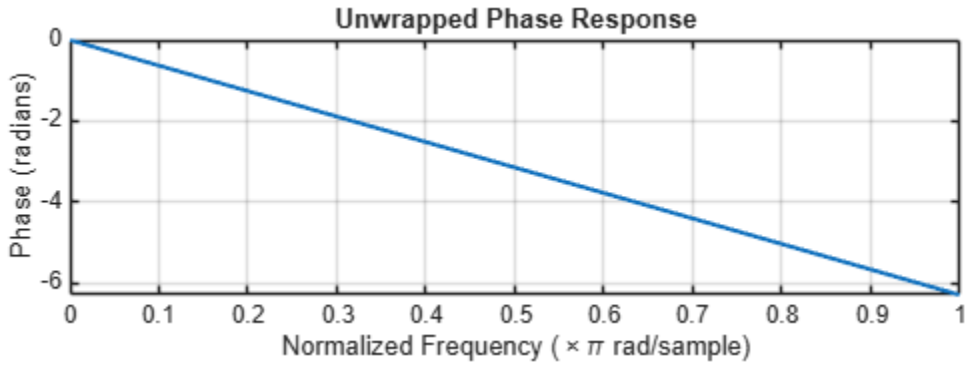
```
figure;

plot(w/pi,phase,'LineWidth',1.5);

grid on;

xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('Phase (radians)');

title('Unwrapped Phase Response');
```



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GRAPH 4: GROUP DELAY ==========

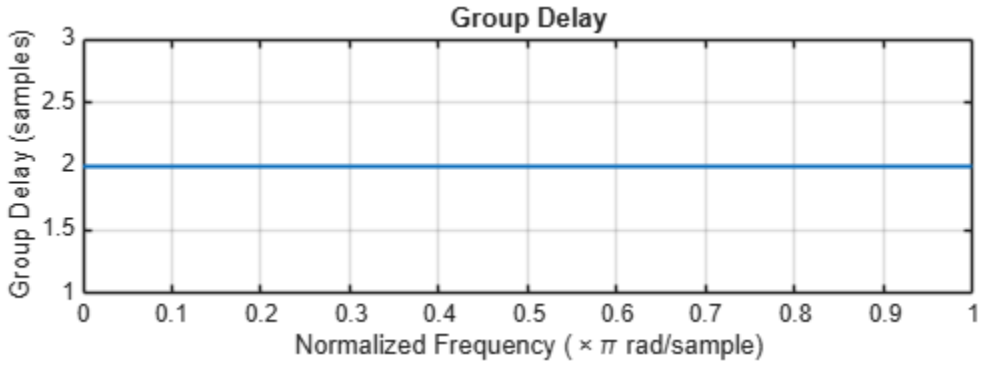
```
figure;

plot(w/pi,gd,'LineWidth',1.5);

grid on;

xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('Group Delay (samples)');

title('Group Delay');
```



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GRAPH 5: NUMERICAL ERROR ==========

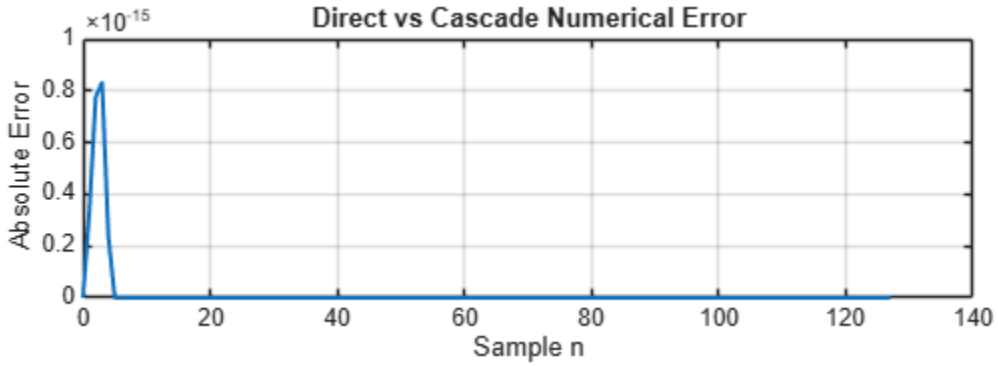
```
figure;

plot(n,abs(yd-yc),'LineWidth',1.5);

grid on;
```

```
xlabel('Sample n');
ylabel('Absolute Error');

title('Direct vs Cascade Numerical Error');
```



==========

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FINAL RESULT ==========

```
fprintf('\n==========================================\n');
fprintf('FINAL RESULT\n');
fprintf('==========================================\n');

fprintf('1. FIR filter realized using Direct structure.\n');
fprintf('2. FIR filter realized using Cascade structure.\n');
fprintf('3. Coefficients are %s.\n',symmetry);
fprintf('4. Linear-phase group delay = %.2f samples.\n',D);
fprintf('5. Maximum realization error = %.3e.\n',err);
fprintf('6. Percentage deviation = %.3e %%.\n',dev);
fprintf('7. Direct and Cascade responses are equivalent.\n');
```

```
==========================================
FINAL RESULT
==========================================
1. FIR filter realized using Direct structure.
2. FIR filter realized using Cascade structure.
3. Coefficients are Symmetric.
4. Linear-phase group delay = 2.00 samples.
5. Maximum realization error = 8.327e-16.
6. Percentage deviation = 2.220e-13 %.
7. Direct and Cascade responses are equivalent.
```

Published with MATLAB® R2026b